

Theoretical Ecotoxicology

Session 4: Dose-response modelling

University Duisburg-Essen SS 2026

Ralf B. Schäfer

Ecotoxicology, Faculty of Biology, UDE & RC One Health Ruhr

Intended learning outcomes of today's lecture

- Understanding the background of dose-response modelling
- Knowledge of different models and steps of model fitting
- Ability to fit models in R

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- Understanding the background of dose-response modelling
 - What is a dose and a concentration?
 - Why should you know about dose response modelling as an ecotoxicologist?
- Knowledge of different models and steps of model fitting
 - Describe the different steps of model fitting.
 - Interpret the parameters of the 4-parameter log-logistic model.
 - How can you evaluate if model assumptions are met and compare which model is most plausible given the data?

Today

- Introduction to dose-response modelling
- From NOECs to NECs
- Species sensitivity distributions
- Steps in dose-response modelling
- Model evaluation and comparison

What is a dose?

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What is a dose (of a chemical)?

- External dose: Amount administered to an organism
 - Exposure at organism level, more common in terrestrial ecotoxicology and toxicology
 - Expressed per body weight: mg/kg
- Internal dose: Amount available determined by toxicokinetics
 - Toxicokinetics include absorption, distribution, metabolism, and excretion of the chemical
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For simplicity, dose in the context of modelling can refer to all these specific doses

What is the difference between a dose and a concentration?

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Dose	Concentration
Amount in organism	Amount in environment
Typically expressed per body weight (e.g. mg/kg body weight)	Typically expressed per volume or mass of medium (e.g. mg/L in water, mg/kg in soil)
Exposure at organism level	Environmental exposure



Dose $\neq$ Concentration, but dose-response curve more commonly used than concentration-effect curve

Typical responses

- Mortality/lethality
- Wide range of sublethal organismal responses
 - Morphology, growth, reproduction, development
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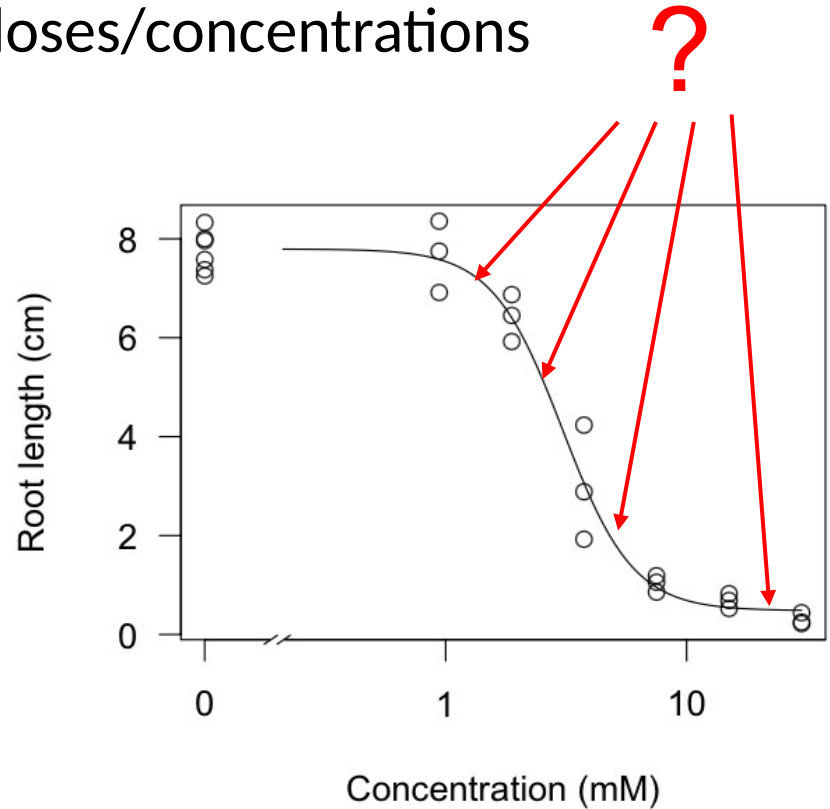


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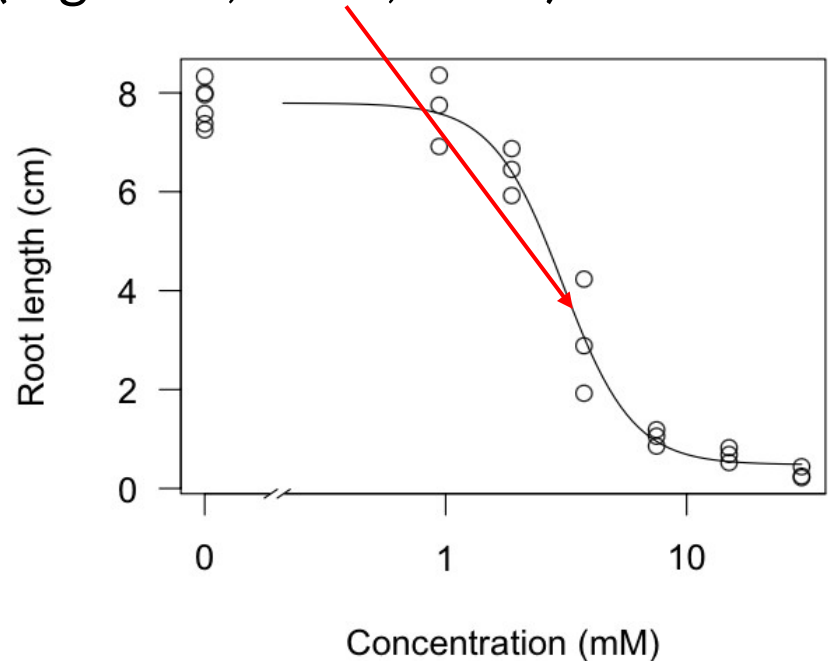
Why do we need dose-response modelling?

1) Predict toxicity at untested doses/concentrations



Why do we need dose-response modelling?

- 1) Predict toxicity at untested doses/concentrations
- 2) Derive regulatory threshold (e.g. NEC, EC50, BMD)



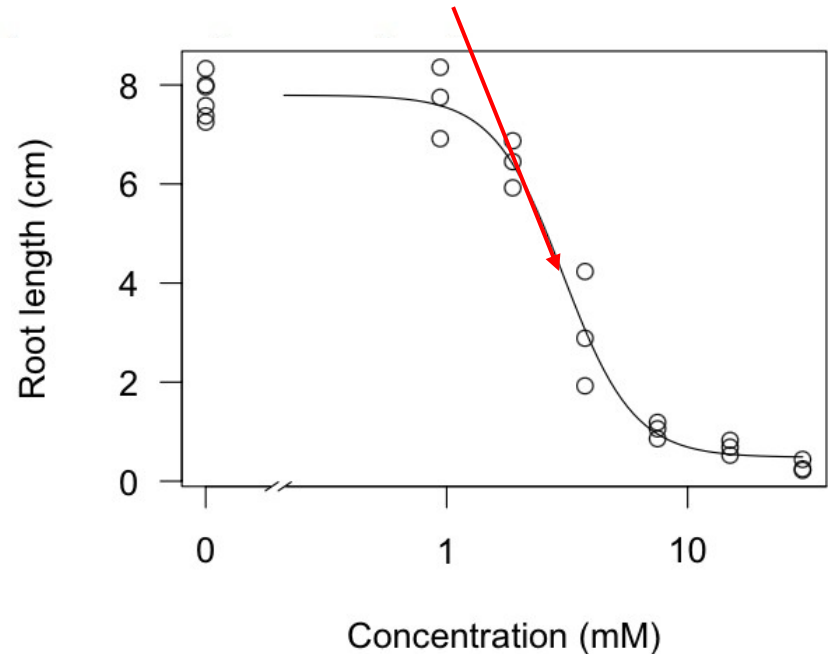
Model-based thresholds

LC₅₀ — The concentration that causes mortality in 50% of the organisms tested estimated by graphical or computational means.

EC₅₀ — The concentration that has an effect on 50% of the organisms tested estimated by graphical or computational means. Often this parameter is used for effects that are not death.

**Model-based
parameters**

Landis, Sofield & Yu 2017: 82



Hypothesis testing-based thresholds

NOEC—No observed effects concentration determined by hypothesis testing.

NOEL—No observed effects level determined by hypothesis testing methods. This parameter is reported as a dose.

NOAEC—No observed adverse effects concentration determined by statistical hypothesis testing methods. The effect is usually chosen for its impact upon the species tested.

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LOEC—Lowest observed effects concentration determined by hypothesis testing methods.

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MTC—Minimum threshold concentration determined by statistical hypothesis testing methods.

MATC—Maximum allowable toxicant concentration determined by graphical or statistical methods.

Derived using ANOVA-type statistics → hypothesis testing

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Example:

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LC ₅₀	1921
(95% confidence intervals) ($\mu\text{g L}^{-1}$)	(1692-2136)

NOEC

LOEC

*Significantly different from the control ($p \leq 0.05$)

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LC ₅₀	1921
(95% confidence intervals) ($\mu\text{g L}^{-1}$)	(1692-2136)

NOEC

LOEC

If only 10, 33, 100, 330, 1000, 3300 and 10000 was tested:

NOEC = 330

LOEC = 1000

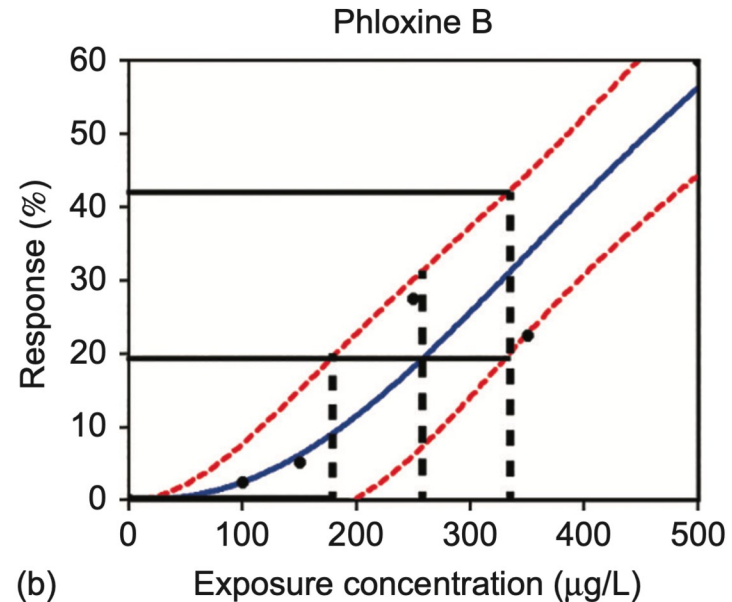
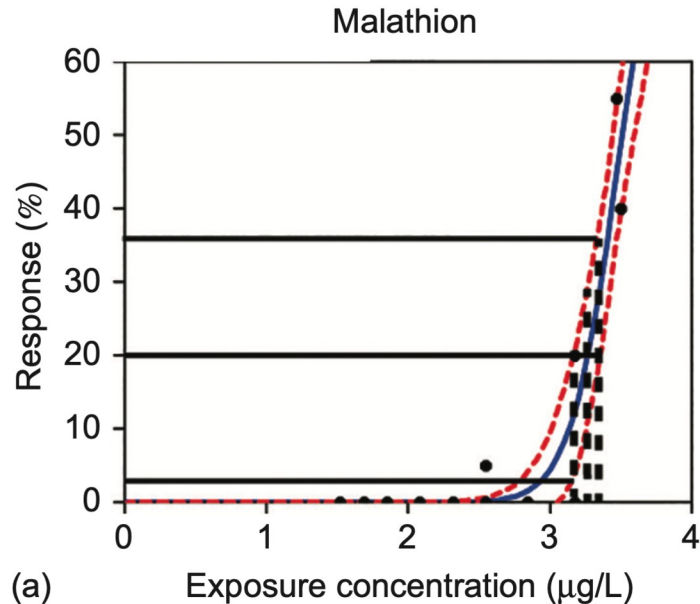
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Disadvantages of hypothesis-based thresholds

- Linked to statistical, not biological significance
 - e.g. 20% deviation from control may not be statistically significant due to insufficient statistical power, but biologically relevant
- Sensitive to experimental design
 - e.g. sample size and replication, number of concentrations tested
- No information on variability of threshold and assessing plausibility requires dose-response relationship

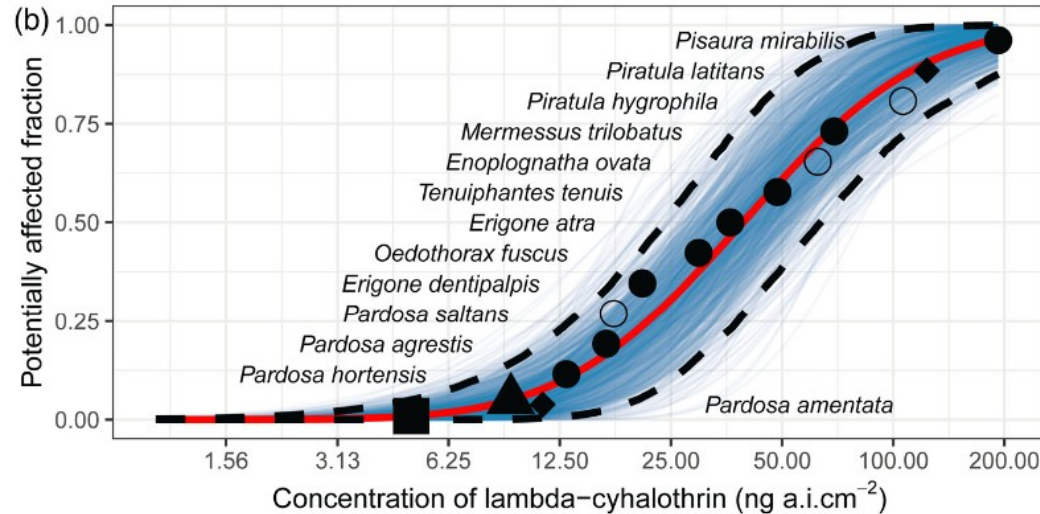
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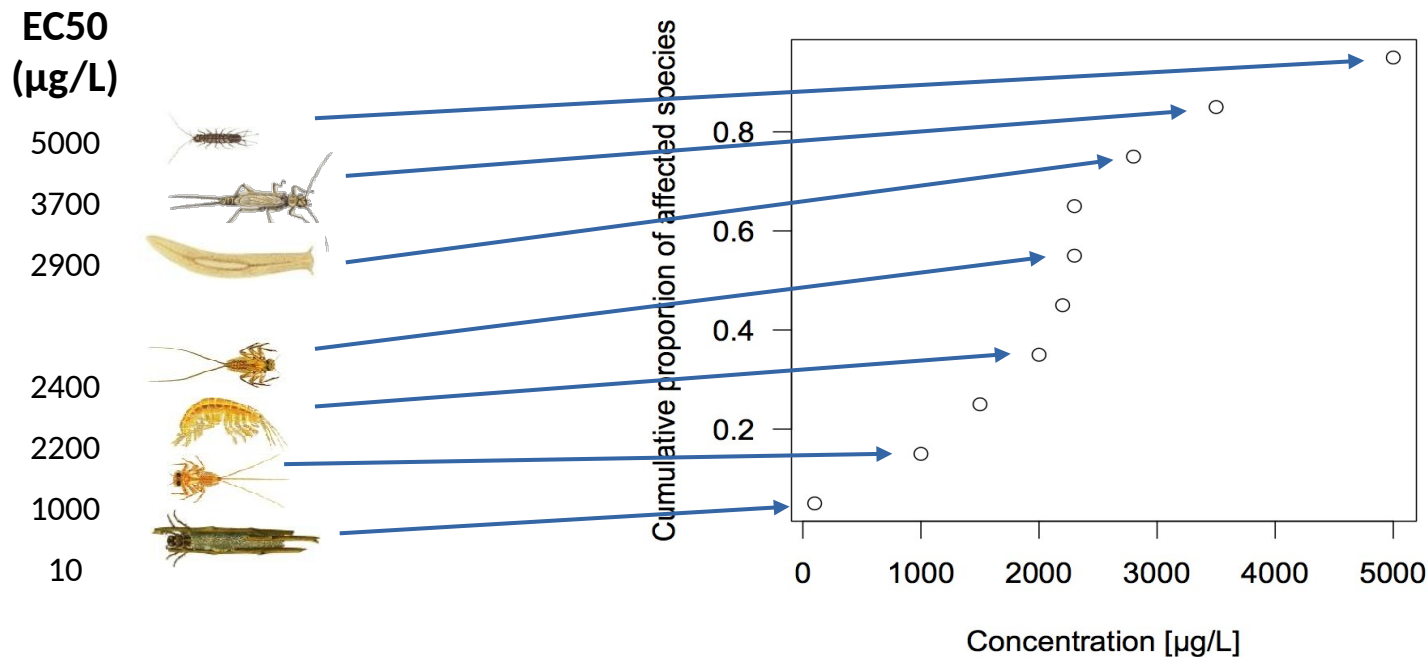
Why do we need dose-response modelling?

- 1) Predict toxicity at untested doses/concentrations
- 2) Derive regulatory threshold (e.g. NEC, EC50, BMD)
- 3) Establish sensitivity distributions for chemicals or species



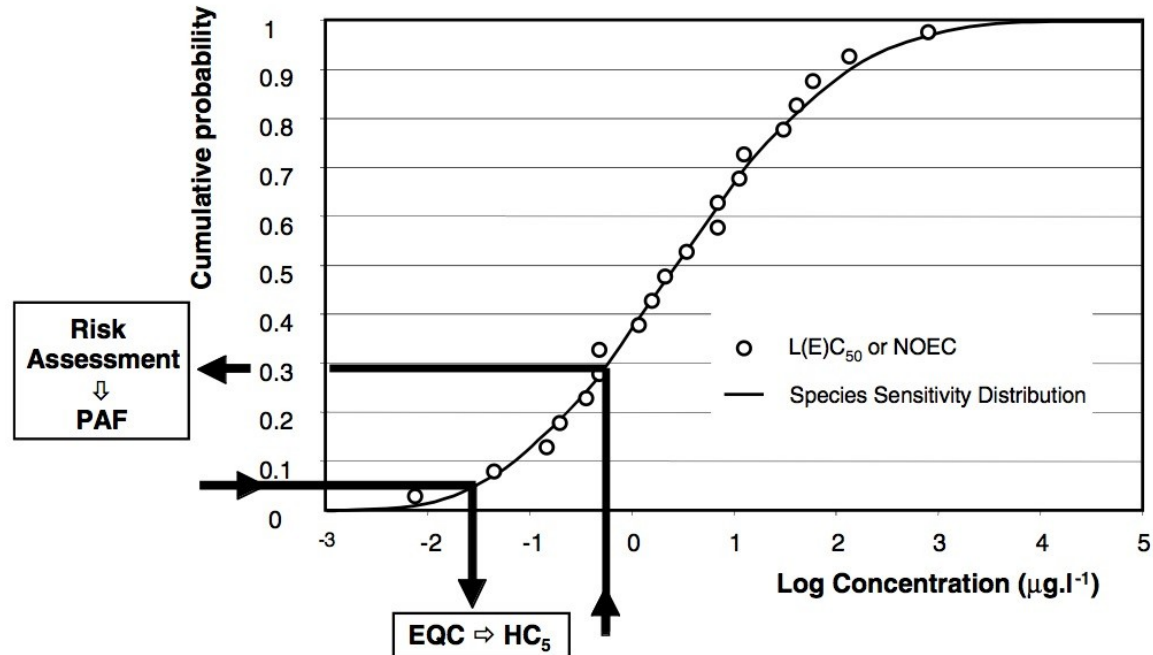
Species sensitivity distributions (SSDs)

Statistical distribution of EC_x (but also NOEC or LOEC) for any chemical within species assemblages



SSDs for risk assessment

- Hazardous concentration (HC_p): conc. at which $p\%$ affected
- Potentially affected fraction PAF: given by concentration



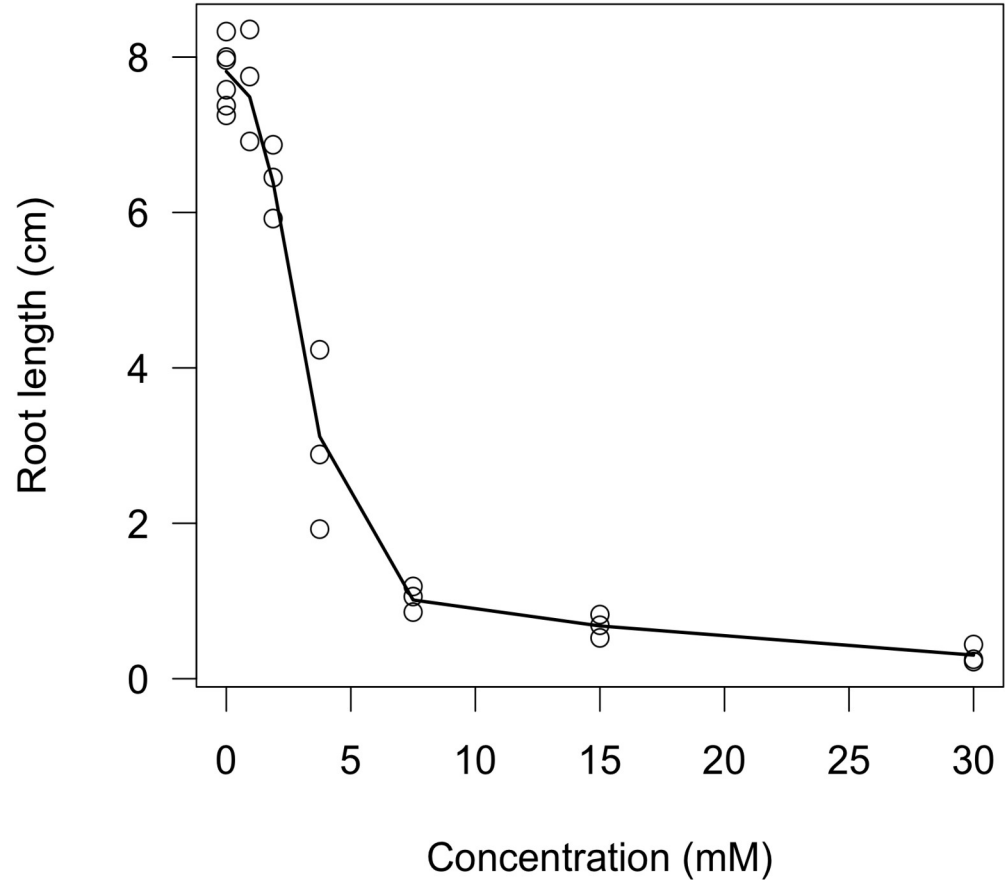
Cutoff for community PNEC

Steps in fitting dose-response models

- 1) Model selection: Select type(s) of model based on characteristics of data (e.g. distribution, replication)
- 2) Fit model(s)
- 3) Evaluate model assumptions (and potentially compare models)
- 4) Derive parameters (e.g. EC_{50})

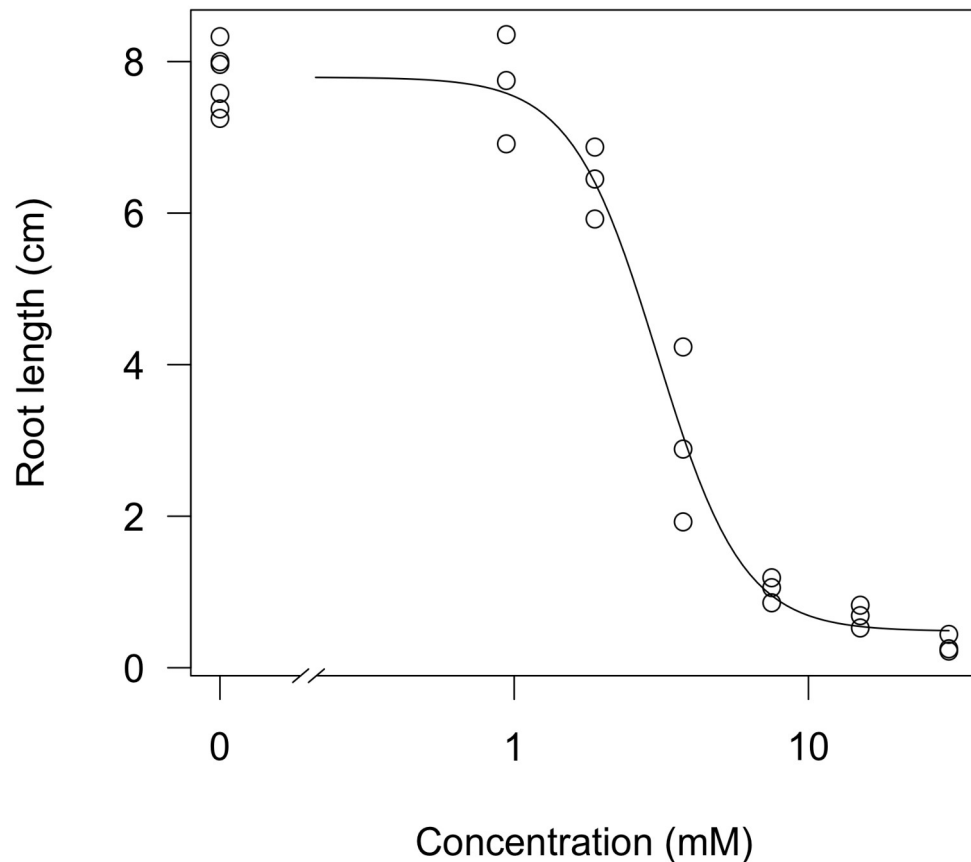
Model selection

Is this how you imagine a
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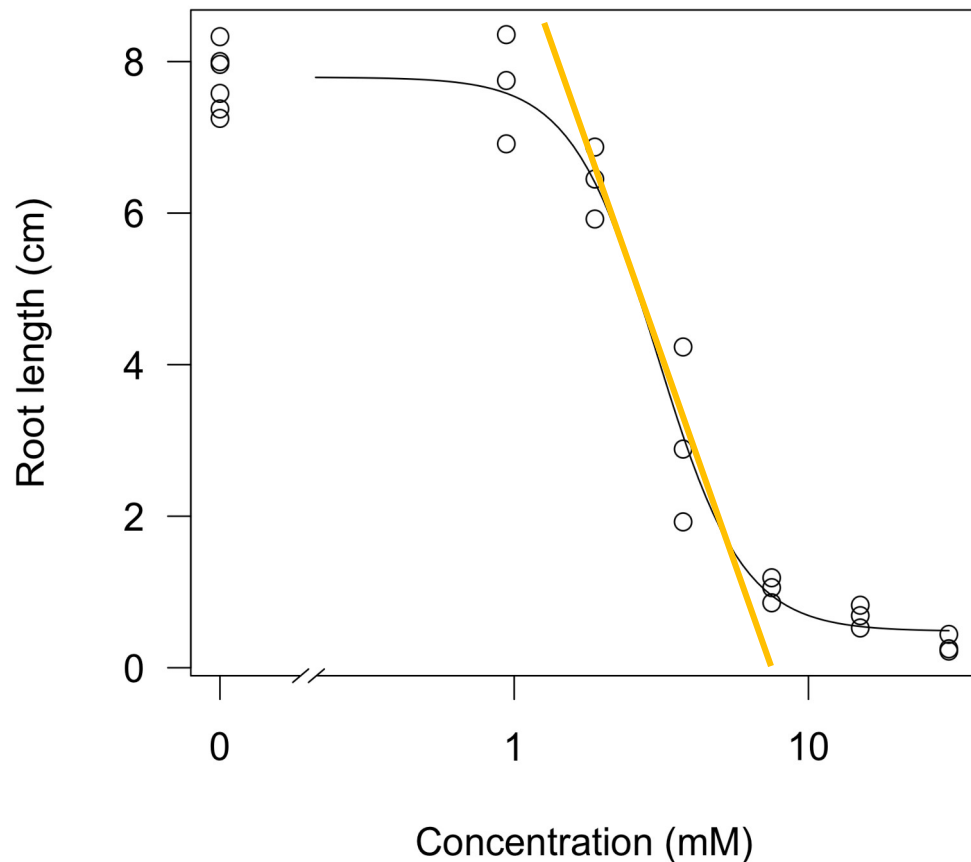
Model selection

- Relationship typically expressed on log-scale, because:
 - Stronger focus on middle range



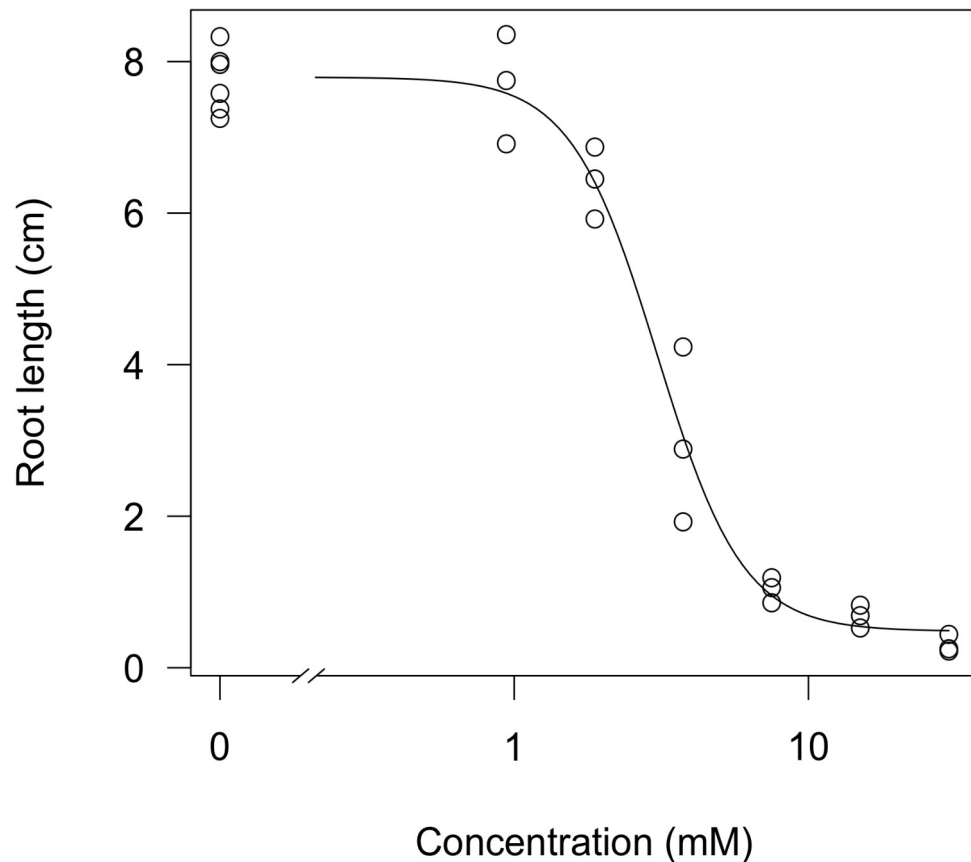
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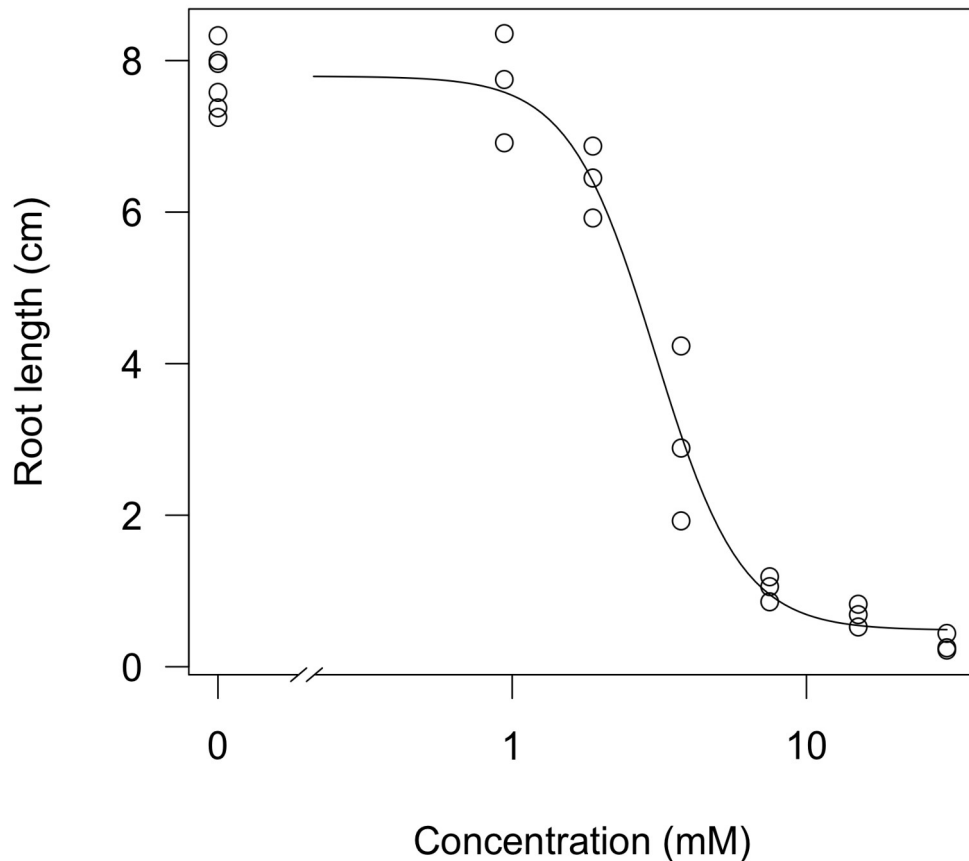
Model selection

- Relationship typically expressed on log-scale, because:
 - Stronger focus on middle range
 - Linearisation of middle portion
 - Less variability in estimation of inflection point (e.g. EC_{50})
 - Comparison of chemicals across wide conc. range

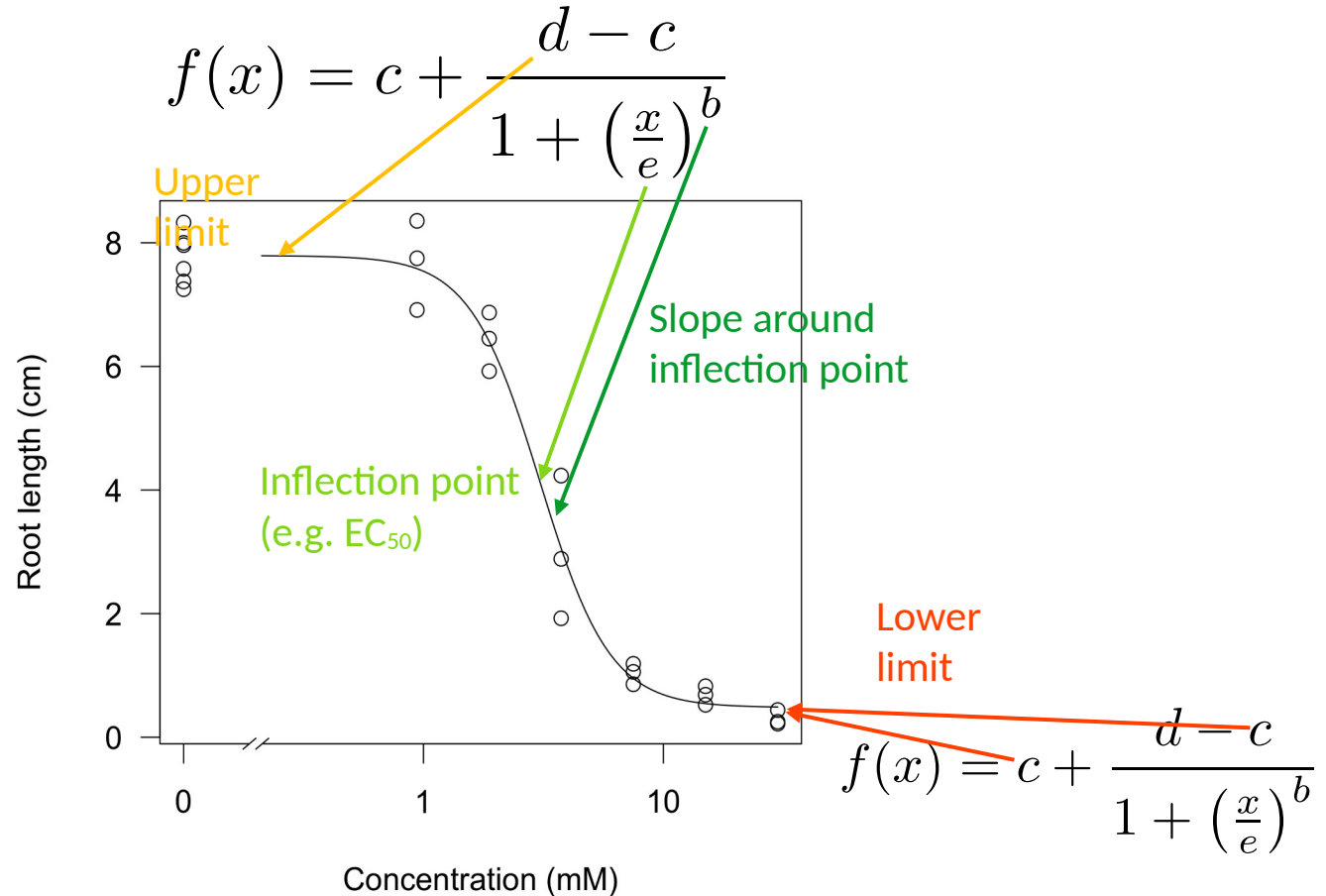


4-parameter Log-logistic model

- Symmetric
- Typically produces reasonable fits, endorsed in guidelines (e.g. from EFSA)
- Parameters with biolog. meaning
- Turns into logit regression for binary data (with max = 1 and min = 0)

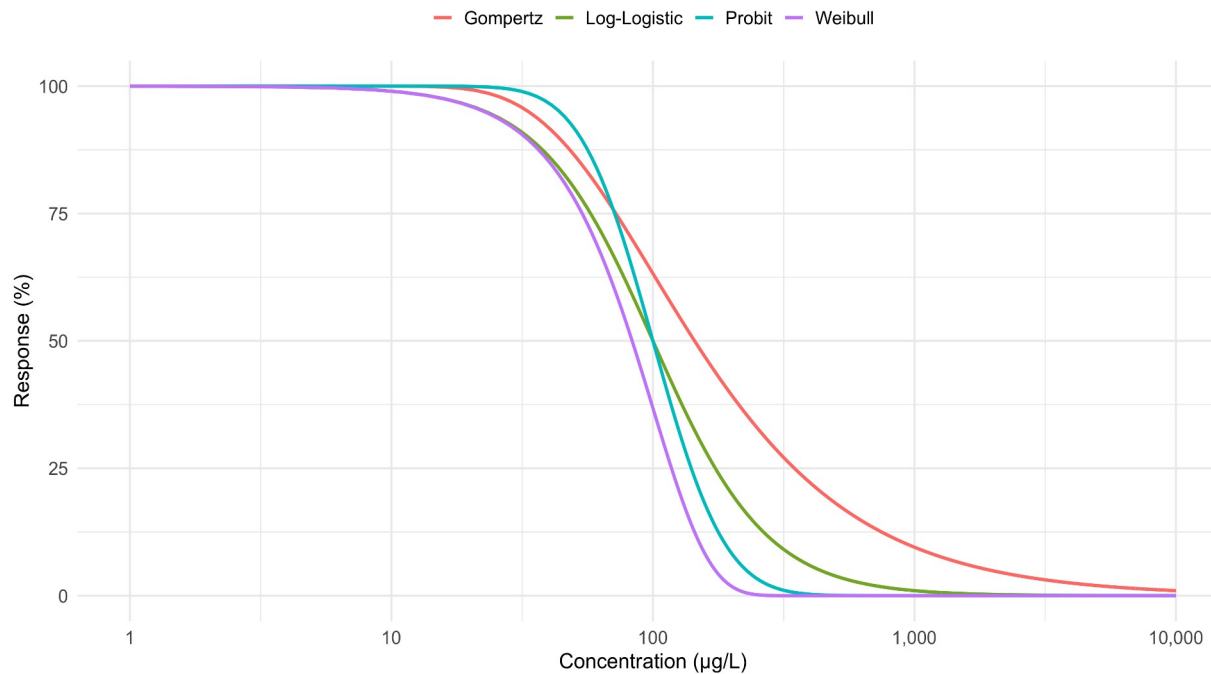


4-parameter Log-logistic model



Model selection

- Diverse range of models
 - Weibull: can model concave and sigmoidal curves; asymmetric; typically used for skewed response
 - Probit: typically used for binary responses; similar to log-normal model
 - Gompertz: typically used for growth and mortality data; slow increase at low dose



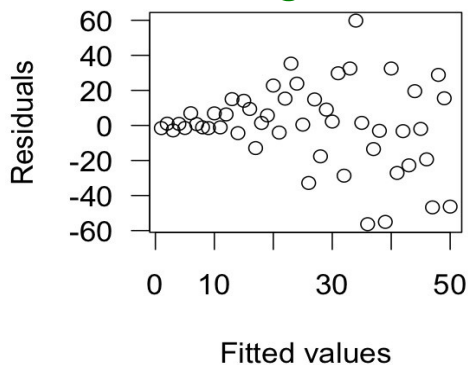
Evaluate model assumptions

- Normal distribution of error → Quantile-Quantile plot
- Variance homogeneity → Residuals vs. fitted values plot
- Outliers → Multiple methods (e.g. Cook's distance)

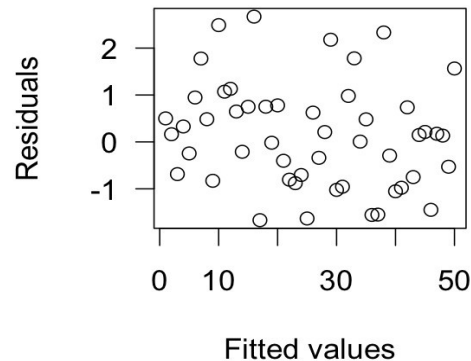
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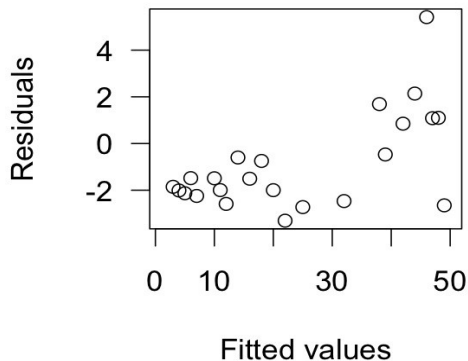
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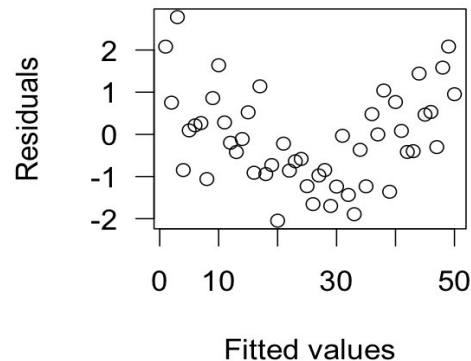
- normal



- slight increase

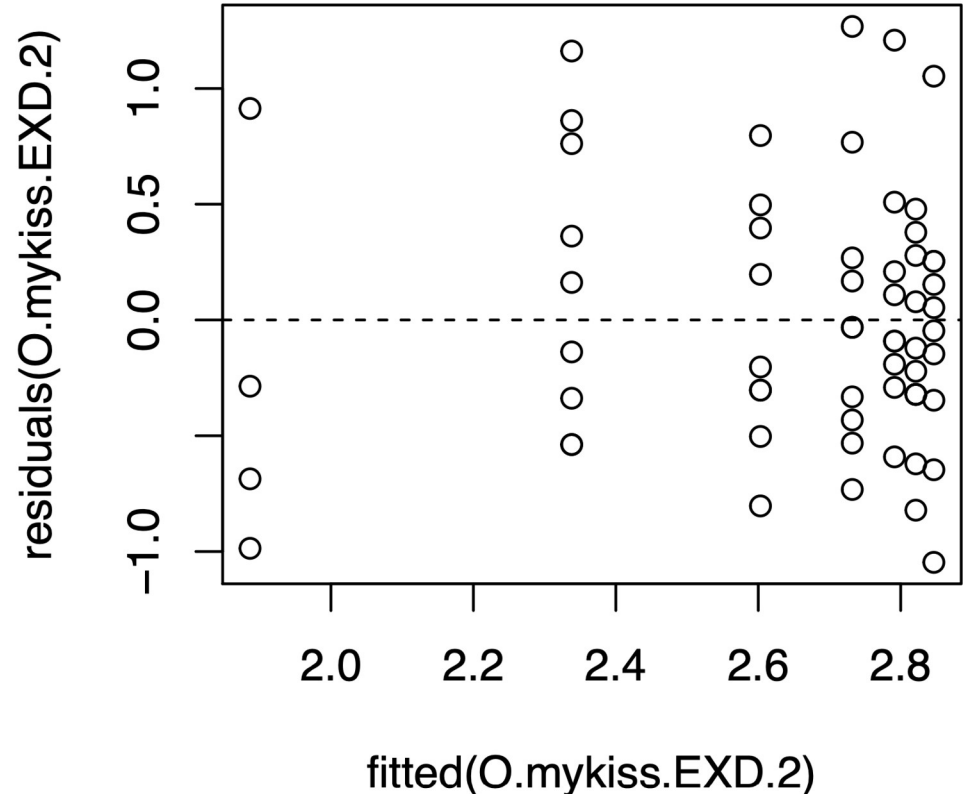


- non-linear



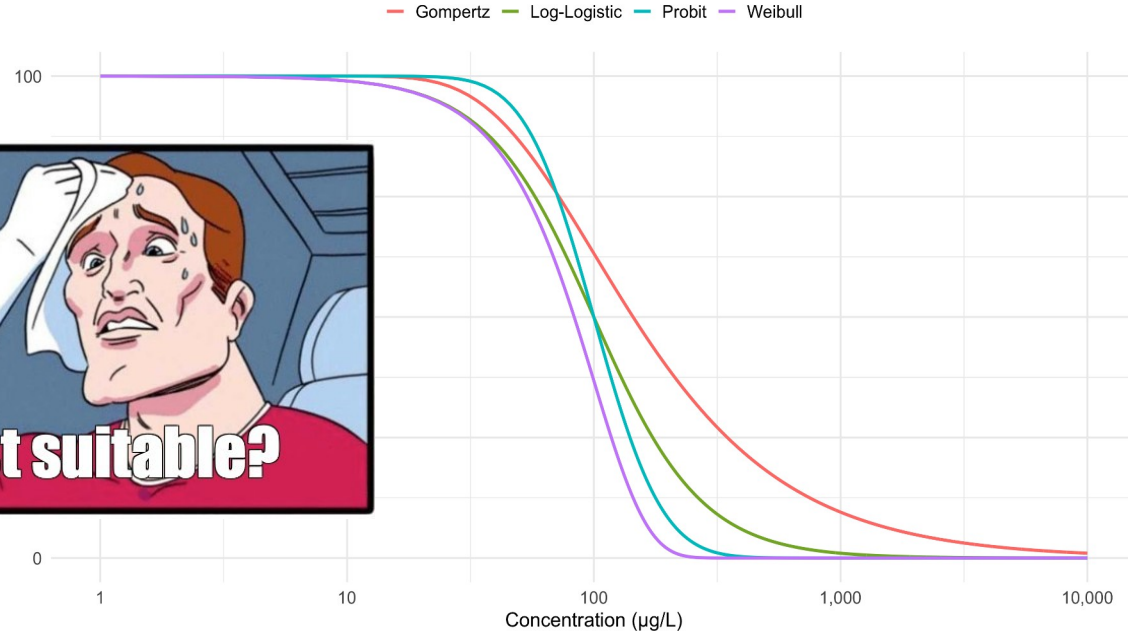
Evaluate variance homogeneity

- Residuals vs. fitted values plots
- Should not show any conspicuous pattern
- For dose response models, typically limited to few levels of fitted values



Compare models

- Goodness-of-fit measures (e.g. Akaike Information Criterion (AIC))
- Confidence intervals (CI)



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$$AIC = -2 \log(L) + 2p$$

L : Likelihood function for the parameters in the model. p : Number of parameters in model

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“It is futile to do with more things that which can be done with fewer”

OCCAM'S RAZOR

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WHO 2009 *Principles for modelling dose-response for the risk assessment of chemicals*: 23.

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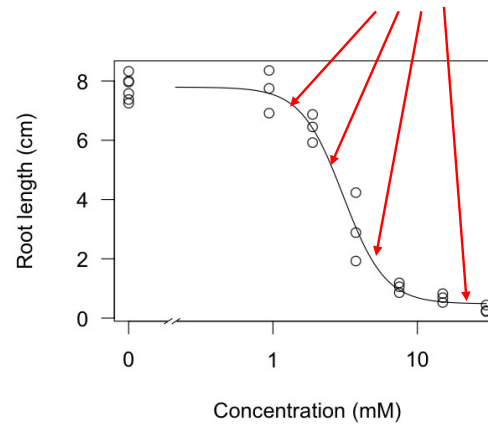
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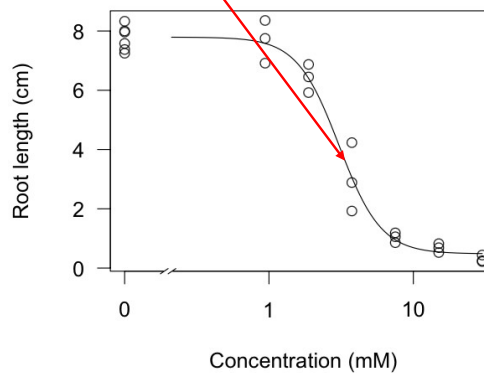
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BMD = Benchmark dose

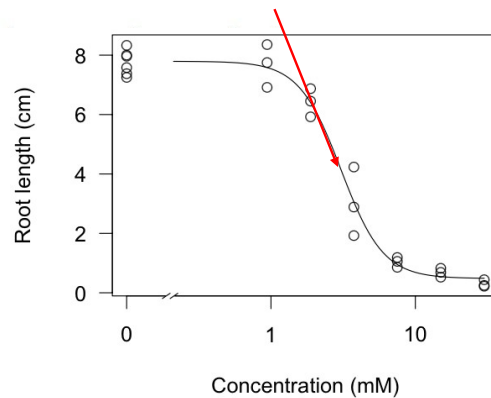
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Landis, Sofield & Yu 2017: 82



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Disadvantages of hypothesis-based thresholds

- Linked to statistical, not biological significance
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Hypothesis-based thresholds such as NOECs have been debated since decades:

Laskowski, R. (1995). Some good reasons to ban the use of NOEC, LOEC and related concepts in ecotoxicology. *Oikos*, 73(1), 140–144.

Jager, T. (2011). Some good reasons to ban ECx and related concepts in ecotoxicology. *Environmental Science & Technology*, 45(19), 8180–8181. <https://doi.org/10.1021/es2030559>

Jager, T. (2012). Bad habits die hard: The NOEC's persistence reflects poorly on ecotoxicology. *Environmental Toxicology and Chemistry*, 31(2), 228–229. <https://doi.org/10.1002/etc.746>

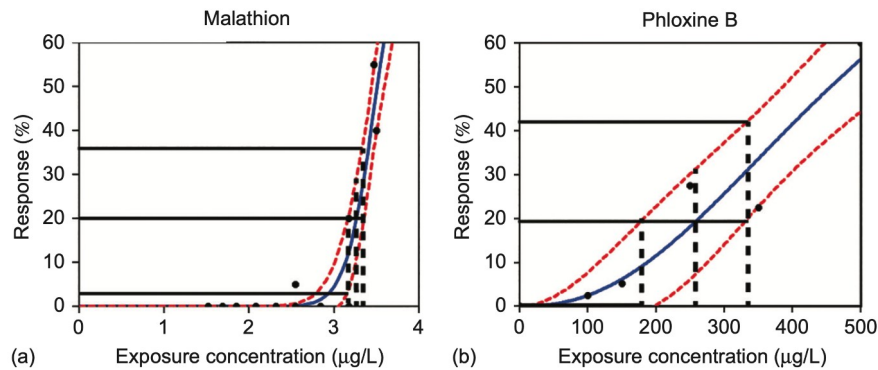
Green, J.W., T.A. Springer, and J.P. Staveley. 2013. The drive to ban the NOEC/LOEC in favor of ECx is misguided and misinformed. *Integr. Environ. Assess. Manag.* 9:12–16

Fox, D. R., & Landis, W. G. (2016). Don't be fooled—A no-observed-effect concentration is no substitute for a poor concentration-response experiment. *Environmental Toxicology and Chemistry*, 35(9), 2141–2148. <https://doi.org/10.1002/etc.3459>

Fisher, R., Fox, D. R., Negri, A. P., Van Dam, J., Flores, F., & Koppel, D. (2024). Methods for estimating no-effect toxicity concentrations in ecotoxicology. *Integrated Environmental Assessment and Management*, 20(1), 279–293. <https://doi.org/10.1002/ieam.4809>

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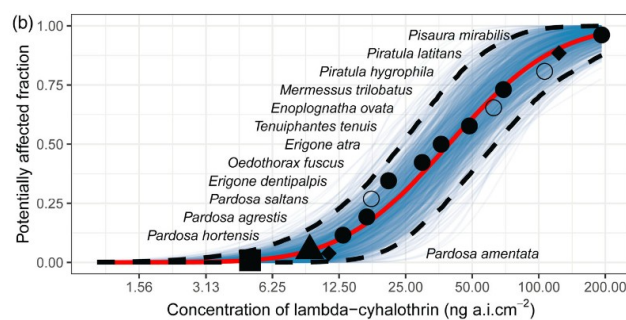


Landis, Sofield & Yu 2017: 88

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Why do we need dose-response modelling?

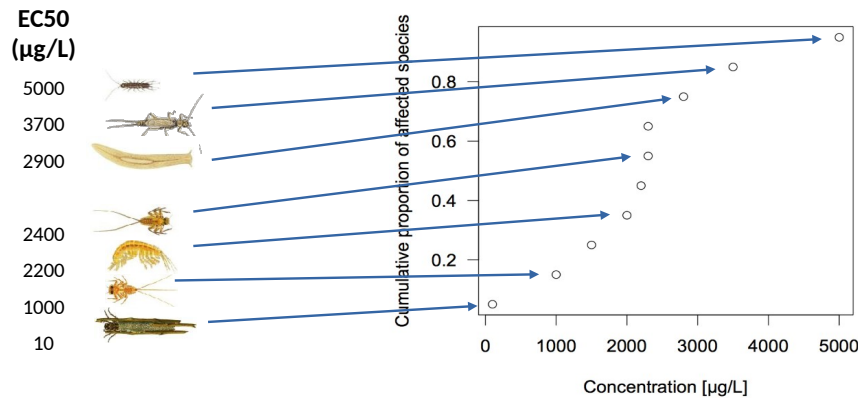
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Duque et al. 2024 *Pest Manag.Sci.* 80

Species sensitivity distributions (SSDs)

Statistical distribution of EC_x (but also NOEC or LOEC) for any chemical within species assemblages



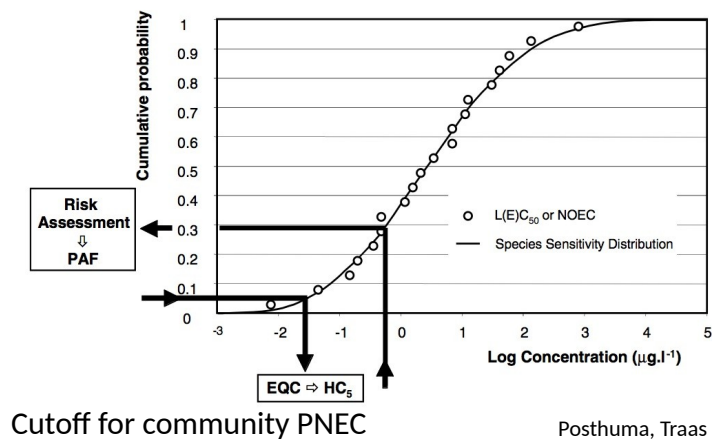
The example is for 10 species. Note that only 7 are shown. Also note that despite the term “assemblage” or “community”, the toxicity data are typically from single species tests.

It is beyond scope to discuss SSDs in depth, but note that it relies on multiple assumptions:

- Species interactions do not influence the SSD
 - Loss of any species is of equal importance
 - Unbiased sample of the target group of species about which conclusions are to be drawn
 - Endpoint (e.g. HC5) is ecologically relevant
 - Chosen level of protection, number of available species, confidence interval and assumed distribution appropriate
- (Forbes & Calow 2002 *Hum. Ecol. Risk Assess.* 8: 473)

SSDs for risk assessment

- Hazardous concentration (HC_p): conc. at which $p\%$ affected
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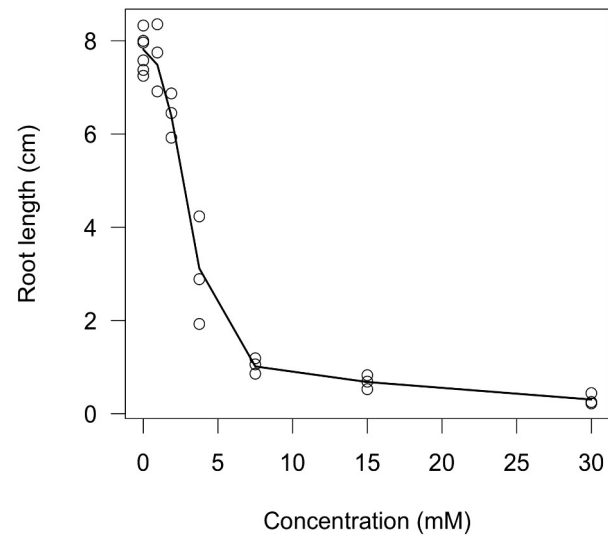
For further discussion of SSDs see: Del Signore, A., Hendriks, A. J., Lenders, H. J. R., Leuven, R. S. E. W., & Breure, A. M. (2016). Development and application of the SSD approach in scientific case studies for ecological risk assessment. *Environmental Toxicology and Chemistry*, 35(9), 2149–2161. <https://doi.org/10.1002/etc.3474>

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- 1) Model selection: Select type(s) of model based on characteristics of data (e.g. distribution, replication)
- 2) Fit model(s)
- 3) Evaluate model assumptions (and potentially compare models)
- 4) Derive parameters (e.g. EC_{50})

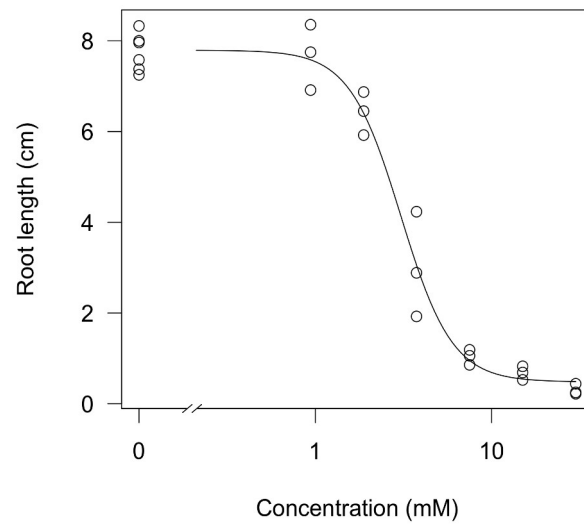
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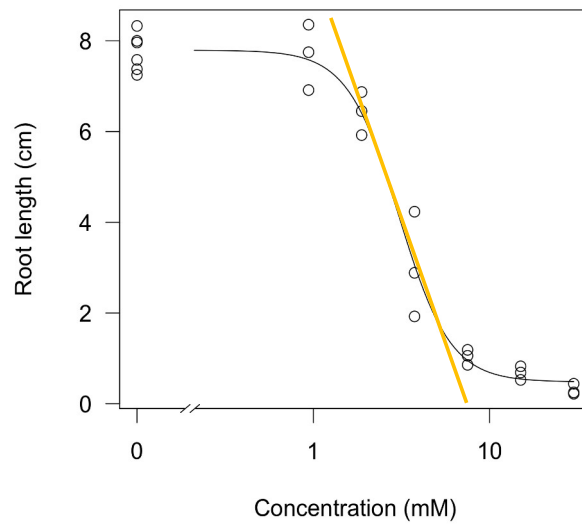
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- Relationship typically expressed on log-scale, because:
- Stronger focus on middle range



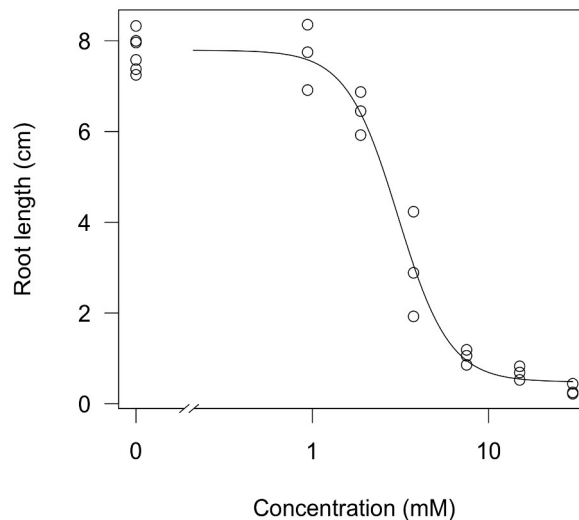
Model selection

- Relationship typically expressed on log-scale, because:
 - Stronger focus on middle range
 - Linearisation of middle portion



Model selection

- Relationship typically expressed on log-scale, because:
 - Stronger focus on middle range
 - Linearisation of middle portion
 - Less variability in estimation of inflection point (e.g. EC_{50})
 - Comparison of chemicals across wide conc. range



But why are biological responses often log-normally distributed?

In simple terms, the law of mass action states that the reaction rate is proportional to the concentrations of the involved reactants. Toxicological effects involve multiple biochemical reactions, which thereby integrate multiplicatively. For example, a signalling cascade with steps $X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow \dots \rightarrow X_n$ amplifies signals multiplicatively:

$$X_n = k_1 \times k_2 \times k_3 \times \dots \times k_{n-1} \times X_1$$

The logarithm of this product is:

$$\log(X_n) = \log(k_1) + \log(k_2) + \log(k_3) \dots + \log(k_{n-1}) + \log(X_1)$$

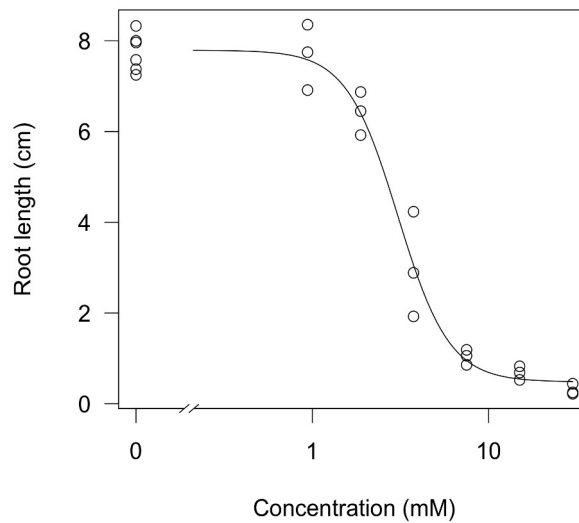
According to the Central Limit Theorem, sums of independent variables tend towards normality. Thus, $\log(X_n)$ tends towards normal distribution, from which follows that X_n is log-normally distributed.

Similarly, cellular thresholds involve multiple processes and variables (e.g. receptor density, enzyme activity, signalling molecule concentrations), which integrate multiplicatively.

However, the log-normal distribution may also be considered biologically advantageous: It handles a wide range of concentrations and is sensitive at very low concentrations, potentially allowing to respond to early warning signals, while avoiding fast saturation.

4-parameter Log-logistic model

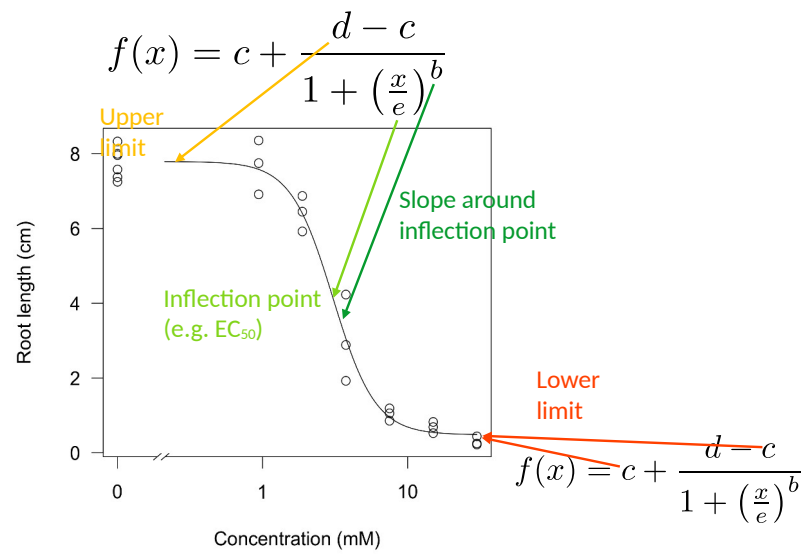
- Symmetric
- Typically produces reasonable fits, endorsed in guidelines (e.g. from EFSA)
- Parameters with biolog. meaning
- Turns into logit regression for binary data (with max = 1 and min = 0)



In a recent comparative study, the log-logistic and log-normal models performed similarly well and better than the Weibull model when fitted for multiple data sets.

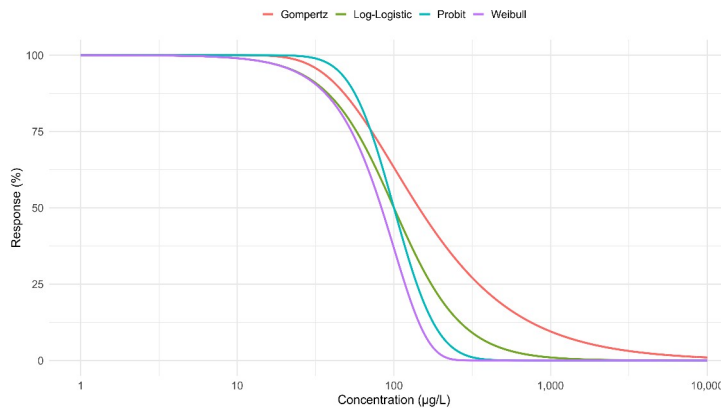
Iwasaki, Y., & Yanagihara, M. (2025). Comparison of model-averaging and single-distribution approaches to estimating species sensitivity distributions and hazardous concentrations for 5% of species. *Environ. Toxicol. Chem.*, 44(3), 834–840. <https://doi.org/10.1093/etjnl/vgae060>

4-parameter Log-logistic model



Model selection

- Diverse range of models
 - Weibull: can model concave and sigmoidal curves; asymmetric; typically used for skewed response
 - Probit: typically used for binary responses; similar to log-normal model
 - Gompertz: typically used for growth and mortality data; slow increase at low dose



For details on these models and how to implement them in R see:

Ritz, C., Jensen, S. M., Gerhard, D., & Streibig, J. C. (2021). Dose-response analysis using R. CRC Press.

Ritz, C., Baty, F., Streibig, J. C., & Gerhard, D. (2016). Dose-Response Analysis Using R. PLoS ONE, 10(12), 1–13. <https://doi.org/10.1371/journal.pone.0146021>

Ritz, C. (2010). Toward a unified approach to dose-response modeling in ecotoxicology. Environmental Toxicology and Chemistry, 29(1), 220–229. <https://doi.org/10.1002/etc.7>

Evaluate model assumptions

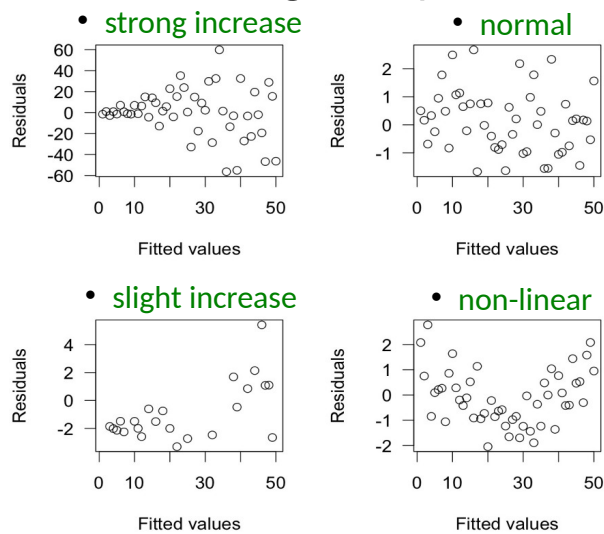
- Normal distribution of error → Quantile-Quantile plot
- Variance homogeneity → Residuals vs. fitted values plot
- Outliers → Multiple methods (e.g. Cook's distance)

A strong deviation of the errors, which are estimated from the residuals, from the normal distribution in a QQ plot may point to an incorrect model choice or skewed data. I assume that you deal with this in a statistics course. However, we will look at this in the practical part.

We will not discuss the topic of outliers further, because the detection and dealing with outliers is a general topic that applies to most models. Moreover, sample sizes are typically low for dose-response experiments, which decreases the reliability of outlier detection.

Evaluate variance homogeneity

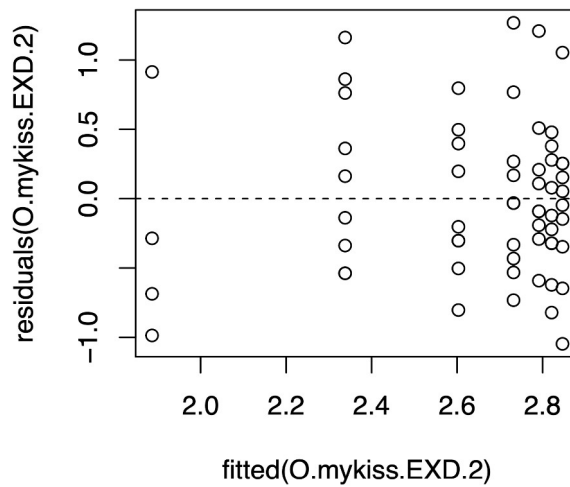
- Residuals vs. fitted values plots
- Should not show any conspicuous pattern



The displayed residuals vs. fitted values plots can be used to check whether the assumption of variance homogeneity holds. If the residuals are not randomly distributed (top right) but display patterns, this may indicate non-constant variance, also termed heterogeneous variance or heteroscedasticity, (bottom and top left) or non-linearity (bottom right).

Evaluate variance homogeneity

- Residuals vs. fitted values plots
- Should not show any conspicuous pattern
- For dose response models, typically limited to few levels of fitted values

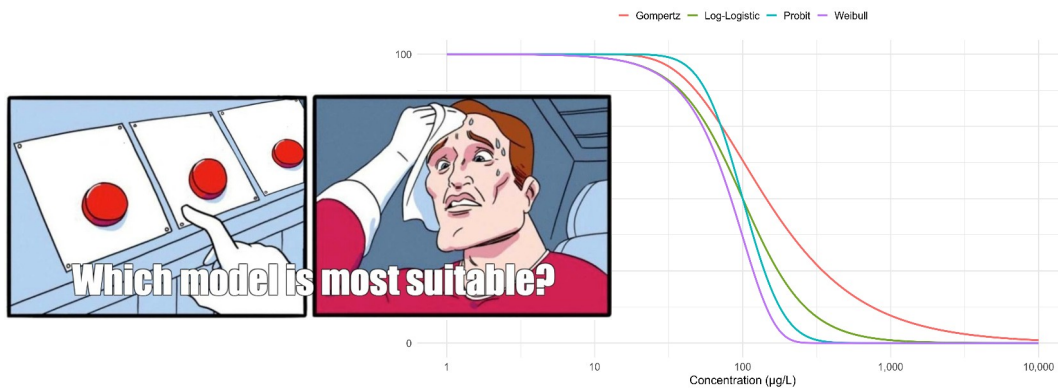


Ritz et al. 2021: 9

Ritz, C., Jensen, S. M., Gerhard, D., & Streibig, J. C. (2021). Dose-response analysis using R (First issued in paperback). CRC Press.

Compare models

- Goodness-of-fit measures (e.g. Akaike Information Criterion (AIC))
- Confidence intervals (CI)



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AIC

$$AIC = -2 \log(L) + 2p$$

L : Likelihood function for the parameters in the model. p : Number of parameters in model

The lower the AIC, the better the model:

- Higher likelihood $\rightarrow$ lower AIC
- Higher number of parameters $\rightarrow$ higher AIC

It is beyond the scope of this course, but if you are not familiar with the likelihood function, you can think of it as a tool that measures how plausible it is that a given set of model parameters would have produced the observed data. The likelihood function increases when the parameters make the observed data more likely under the model. Hence, when comparing models, the model with the highest likelihood would be most suitable.

We will not discuss the use of confidence intervals, as you should be familiar with their interpretation. Otherwise consult a standard statistics book.

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OCCAM'S RAZOR

"It is futile to do with more things that which can be done with fewer"

<http://www.phdcomics.com/comics/archive.php?comid=1237>

Occam's razor may be translated to our situation as:
Given a similar predictive or explanatory power,
models with fewer variables are generally better
than those with more variables.